Instance identity and spatial update diagnosis

Four exposed development sources per direction | all predictions sealed before new GT scoring

Matched arm	HC vIoU	Vid vIoU	HC sIoU	Vid sIoU
Frozen	22.210	14.490	47.895	48.458
Temporal QA	31.286	22.442	47.895	48.458
QA + strict-old IoU OPD	32.436	25.678	44.358	60.679
QA + decoupled IoU OPD	32.436	31.899	44.358	69.942
QA + presence admission	32.436	31.899	44.358	69.942
QA + loss-weighted OPD	32.436	31.899	44.358	69.942
QA + validated OPD	33.601	31.899	47.363	69.942
QA + strict geometry	36.410	26.187	63.631	62.210
QA + decoupled geometry	36.410	34.788	63.631	76.713

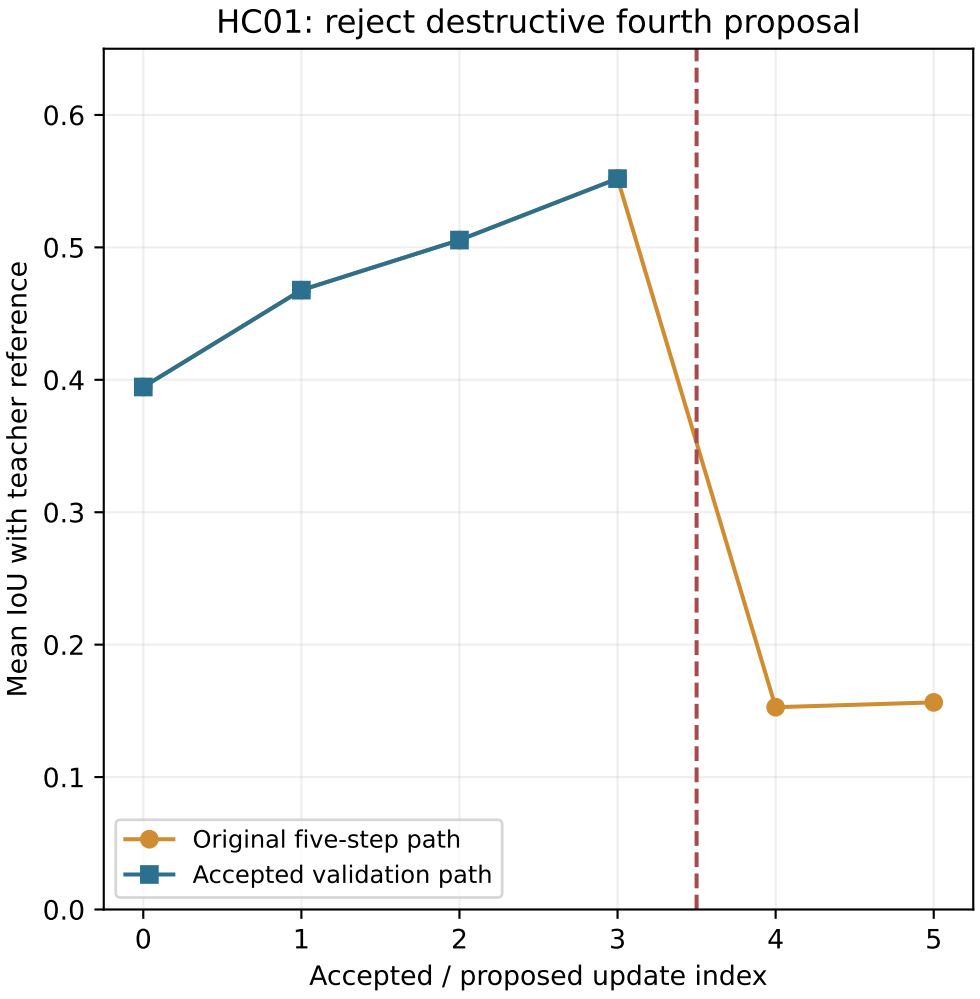
WHO/WHEN separation: V01 and V02 each recover four real mask anchors.

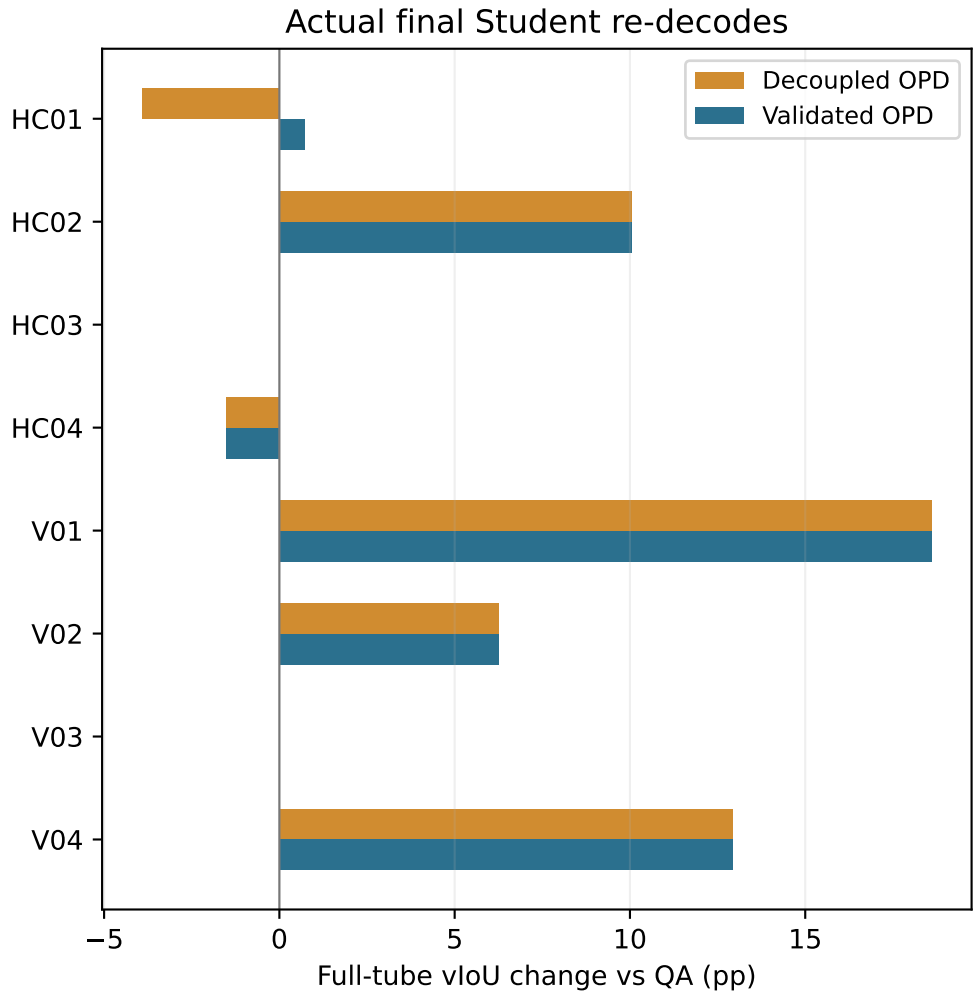
Validated OPD: HC +2.315 pp; VidSTG +9.457 pp relative to Temporal QA.

Geometry-only still leads; HC04 remains negative. Reliability weights lack variation in this panel.

Research candidate only. No production promotion or independent/full-data expansion.

Mechanism evidence and all eight outcomes





Reference agreement is label-free; it is not a semantic-correctness or unseen-frame guarantee.

HC02 and V04 gains are retained. V01/V02 benefit from identity admission; HC04 still declines.

16 new fit paths | 69 Adam proposals | 68 accepted | 1 rejected | 16 final full-model re-decodes